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Highly efficient iridium(III) phosphors with phenoxy-substituted ligands and their high-performance OLEDs†

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Two new iridium(III) cyclometallated complexes (**1** and **2**) based on the 2-(1-phenoxy-4-phenyl)-5-methylpyridine ligand have been developed. By attaching a flexible phenoxy group on the phenyl ring of 2-phenylpyridine (Hppy), the light-emitting properties of the resulting Ir^{III} complexes have been improved, while the introduction of an electron-donating methyl group on the pyridyl ring of Hppy can keep the triplet emission in the green region by compensating for the reduced energy gap caused by the phenoxy group. Owing to the unique electronic structures induced by the ligand, the vacuum-evaporated organic light-emitting devices (OLEDs) of the type [ITO/NPB (40 nm)/(**1** or **2**):CBP (20 nm)/BCP (10 nm)/Alq_3 (30 nm)/LiF (1 nm)/Al (100 nm)] furnished peak OLED efficiencies at 10.0%, 31.1 cd A^{-1} and 14.5 lm W^{-1} for **1** and 11.7%, 38.1 cd A^{-1} and 31.8 lm W^{-1} for **2**. By replacing the electron-injection/electron-transporting materials (BCP and Alq_3) with TPBi, the green-emitting devices based on **1** gave outstanding peak efficiencies at 22.5%, 76.2 cd A^{-1} and 72.8 lm W^{-1}. Extremely high peak efficiencies of 24.5%, 84.6 cd A^{-1} and 77.6 lm W^{-1} were even obtained for the **2**-doped devices and both of them are superior in performance to the benchmark dopants $Ir(ppy)_3$ and $Ir(ppy)_2(acac)$. Moreover, polymer light-emitting devices were also fabricated using **1** and **2** *via* the spin-coating method, and their device performances are characterized by 14.4%, 39.5 cd A^{-1} and 12.4 lm W^{-1} for **1** and 12.6%, 29.6 cd A^{-1} and 18.1 lm W^{-1} for **2**. When **2** was used to make three-color white-light OLEDs, respectable device efficiencies of 15.3 cd A^{-1}, 7.5% and 9.1 lm W^{-1} were achieved and their white color CIE coordinates are improved relative to $Ir(ppy)_3$.

Introduction

Research on organic light-emitting diodes (OLEDs) has shown tremendous progress in the past few decades. The highly efficient light-emitting materials, which are one of the most critical components in the advancement of OLEDs, are always drawing the attention of the scientific community. Since the famous work on a high-efficiency OLED using *fac*-$Ir(ppy)_3$ (Hppy = 2-phenypyridine),[1] much attention has been paid on the area of $Ir(ppy)_3$-based electroluminescent materials.[2] Many researchers modified the Hppy ligand with different functional groups in recent years in order to obtain suitable phosphorescent dyes for full-color flat-panel displays and low-cost lighting sources.[3] Although extensive research on the investigation of functional group modification on Hppy ligand has been well documented in the literature, *fac*-$Ir(ppy)_3$ is still the benchmark green phosphor in the OLED applications owing to its easy synthesis, good stability and high efficiency. Furthermore, the introduction of functional groups to Hppy ligand will change its electronic structure, resulting in different colors for the triplet emission, for example, the emission wavelength of an Ir^{III} complex based on ppy ligand will be red-shifted under the following conditions: (a) an electron-donating group is introduced on the phenyl ring of ppy; (b) the π-conjugation length of the ligand is extended; and (c) an electron-withdrawing group is added to the pyridyl ring of ppy.[4] So, derivatization of the ppy

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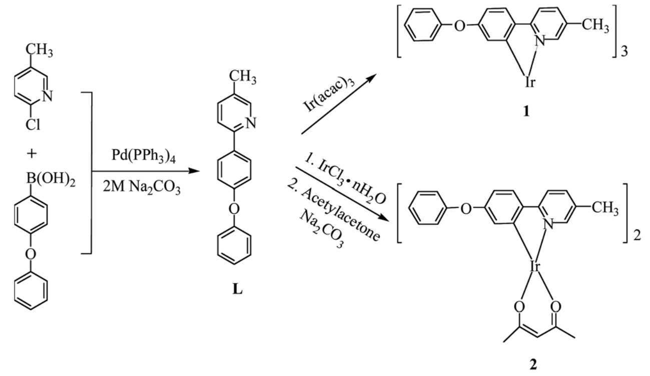
ligand with functional moieties can lead to blue-shift or red-shift of the emission spectra of the resulting Ir^{III} cyclometallates. On the other hand, one of the essential factors for highly efficient electrophosphorescent OLEDs is the compatibility between the phosphorescent dopant and organic host, which requires highly amorphous character of the triplet emitters.[5] Hence, rigid conjugated units should not be used in functionalizing the ppy ligand of green-emitting Ir^{III} phosphors so that the occurrence of crystallization can be avoided during the fabrication.

On this basis, we designed two new Ir^{III} cyclometallated complexes containing the ligand 2-(1-phenoxy-4-phenyl)-5-methylpyridine, which contains a flexible phenoxy group on the phenyl ring and an electron-donating methyl group on the pyridyl ring of Hppy. The phenoxy group at the 4-position of the phenyl ring improves the emission and morphological features of the emission layer in OLED devices due to the presence of the branched and twisted configuration of the substituted main-group unit. This prevents molecules from stacking and induces morphologically stable amorphous thin-film formation, leading to improvement of thermal stability and good performance of electrophosphorescent devices.[6] The phenoxy group is a weak electron-donating group, which would often cause a slight bathochromic shift of the emission wavelength. In keeping the designed Ir^{III} cyclometallates as green phosphors, we have anchored a methyl group at the 5-position of the pyridyl ring, which is a moderate electron-donating group that can help compensate for the narrowing of the energy gap caused by the phenoxy group. Also, the presence of the methyl group enhances the solubility of Ir^{III} cyclometallates and increases their compatibility with the organic host to improve the performance of electroluminescent devices.[7] Here, we describe the results of our investigation on the synthesis, structural characterization, photophysical, redox and thermal properties, theoretical computations and OLED performance of two Ir^{III} cyclometallated complexes containing 2-(1-phenoxy-4-phenyl)-5-methylpyridine. These admirable results indicate the great potential of these two Ir^{III} green phosphors which are as competitive as the state-of-the-art green triplet emitters *fac*-$[Ir(ppy)_3]$ and $Ir(ppy)_2(acac)$ (Hacac = acetylacetone).

Results and discussion

Synthetic strategies and chemical characterization

The chemical structures and synthetic procedure of new Ir^{III} cyclometallates are shown in Scheme 1. The organic cyclometallating ligand **L** was prepared by the Suzuki cross-coupling between 4-phenoxyphenylboronic acid and 2-chloro-5-methylpyridine with an overall yield of 63%. The homoleptic complex **1** was synthesized in a single step from the cyclometallation of $[Ir(acac)_3]$ with **L** in glycerol over *ca.* 200 °C,[8] while the heteroleptic complex **2** was prepared in two steps from the cyclometallation of $IrCl_3 \cdot nH_2O$ with **L** to form the chloride-bridged dimer, followed by reaction with Hacac in the presence of Na_2CO_3.[9] Purification of the products was carried out by silica gel column chromatography or preparative TLC plates using a mixture of dichloromethane and hexane as the eluent to give yellow powders of **1** and **2**.



Scheme 1 Synthesis of new phosphorescent homoleptic and heteroleptic Ir^{III} cyclometallates containing the phenoxy-substituted ppy ligand.

Both of the Ir^{III} cyclometallates **1** and **2** were fully characterized by NMR spectroscopy and fast atom bombardment mass spectrometry (FAB-MS). ^1H and ^{13}C NMR resonance peaks suggested that the stereochemistry of the triply cyclometallated complex **1** is that of the facial isomer with inherent C_3 symmetry because the NMR chemical shifts observed in the three ligands are equivalent. The characteristic singlet resonance peaks located at 1.82 and 5.22 ppm in the ^1H NMR spectra and the resonance peaks at 28.47 and 100.45 ppm in the ^{13}C NMR spectra are ascribed to the auxiliary $acac^-$ anion in **2**. In each case, the FAB-MS spectrum reveals the respective parent ion peak clearly.

Furthermore, the three-dimensional structures of **1** and **2** were also determined by single-crystal X-ray diffraction studies. For the homoleptic complex **1**, the Ir^{III} core adopts a pseudo-octahedral coordination mode with a facial geometry (Fig. 1). Compound **2** possesses a distorted octahedral geometry around the Ir^{III} centre, which consists of two cyclometallated C^N ligands and one $acac^-$ ligand (Fig. 2). The X-ray crystallographic results are consistent with the presence of $Ir(C\hat{}N)_3$ and $Ir(C\hat{}N)_2$ fragments in **1** and **2**, respectively.[10] The detailed structural data are given in ESI (Tables S1 and S2†).

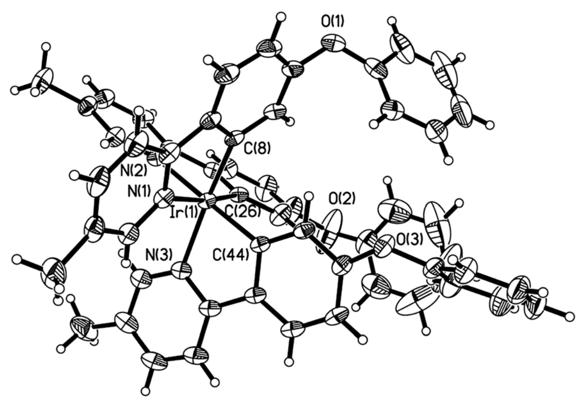


Fig. 1 An ORTEP drawing of **1** with thermal ellipsoids drawn at the 25% probability level. Labels on carbon atoms (except for those bonded to the Ir centre) are omitted for clarity.

Thermal and photophysical properties

The main-group moieties such as arylsilanes are expected to provide high thermal and chemical stability as well as glassy properties when incorporated into the Ir(ppy)$_3$ core.[11] The thermal properties of **1** and **2** were examined by thermogravimetric analysis (TGA) under nitrogen. The TGA data reveal the good thermal stability of **2** with 5% weight-reduction temperature ($\Delta T_{5\%}$) above 356 °C. Furthermore, **1** even showed a better thermal stability with decomposition temperature as high as 403 °C. Such excellent thermal stabilities of **1** and **2** keep them undecomposed during the vacuum deposition process, avoiding decomposition products to contaminate the OLED and cause poor device performance.[12] Analysis of the DSC traces for **1** and **2** showed no crystallization and melting peaks but only glass-transition temperatures (T_g), which can be attributed to the branched configuration of the substituted phenoxy group despite the relatively small ligand size of the complexes. Both of them showed high T_g values (132 °C for **1** and 116 °C for **2**).

The photophysical characterization results for the Ir^{III} cyclometallates are shown in Fig. 3 and Table 1. The absorption spectra can be divided into two regions. The intense bands observed in the ultraviolet region (**1**: <330 nm; **2**: <350 nm) of the spectrum can be safely assigned to the ligand-based spin-allowed ($^1\pi$–π^*) transitions from the conjugated organic ligands.[2a] The lower-energy absorption bands (**1**: >330 nm; **2**: >350 nm) appear with lower log ε values, suggesting that there is a substantial mixing of the ligand-based $^3\pi$–π^* states and spin-forbidden triplet metal-to-ligand charge-transfer (^{3}MLCT) states.[3a] The presence of these weak bands indicates a strong spin–orbit coupling effect between the Ir^{III} centre and ligands, which makes the ^{3}MLCT ← S_0 transition allowed by the significant singlet–triplet state mixing due to the heavy-atom effect of Ir^{III}.[13]

Under the irradiation of UV light at 400 nm, both Ir^{III} cyclometallates emit intense green phosphorescence in CH_2Cl_2 at room temperature. The emission peaks for **1** and **2** are located at 495 and 502 nm, respectively. Owing to the higher ligand field strength of acac$^-$ ion, the emission maximum of **2** is red shifted slightly from that of **1**.[14] Common to other similar complexes, the phosphorescence of Ir^{III} ppy-type phosphors comes from the ^{3}MLCT states formed by the electron transfer from the Ir^{III} metal centre to the ppy system within the ligand. The Ir^{III} phosphors have phosphorescence quantum yields (Φ_P) of 0.70 for **1** and 0.73 for **2**, relative to *fac*-Ir(ppy)$_3$ ($\Phi_P = 0.40$) in degassed CH_2Cl_2 solutions. The Φ_P values of **1** and **2** are much higher than those of *fac*-Ir(ppy)$_3$, indicating that functionalization of ppy with a flexible phenoxy group and a methyl substituent can enhance the phosphorescence efficiency in solution. The observed phosphorescent lifetimes, τ_p, are 2.0 μs (**1**) and 3.0 μs (**2**), which are comparable to those of other reported Ir^{III} phosphors based on the ppy sketch.[15] Such short lifetimes of **1** and **2** can increase spin-state mixing and reduce the chance of device efficiency rolling-off at high current density, because longer τ_p usually results in more significant triplet–triplet annihilation.[16]

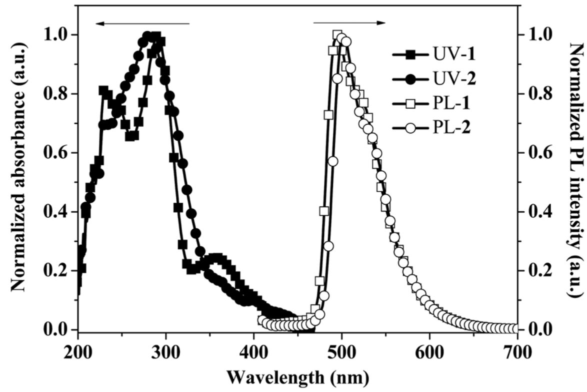


Fig. 3 Absorption and photoluminescence (PL) spectra for the Ir^{III} complexes **1** and **2** in CH_2Cl_2 at 293 K.

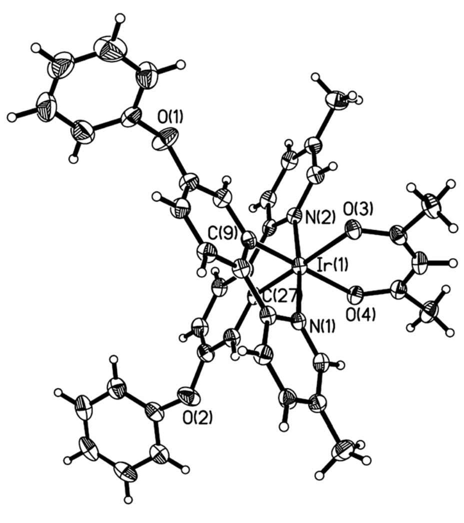


Fig. 2 An ORTEP drawing of **2** with thermal ellipsoids drawn at the 25% probability level. Labels on carbon atoms (except for those bonded to the Ir centre) are omitted for clarity.

Theoretical computations

The nature of the frontier MOs for **1** and **2** has been addressed by means of density functional theory (DFT) and time-dependent density functional theory (TD-DFT) calculations (B3LYP). The contour plots of the frontier MOs for **1** and **2** and the computed electronic density of the frontier orbitals are presented in Fig. 4 and Table S3,† respectively. The HOMO of **1** is mainly composed of the π-system of the ppy core and the Ir atom (mainly d_{xy}). The LUMO also possesses a major contribution from the ppy ligands with a much reduced effect of the metal atom. Only the HOMO and LUMO are described here as they are most relevant to this work. For **2**, the same description may be given except that there is a little contribution of the π-orbital localized on the O atom of the acac$^-$ ligand. Consequently, there are more atomic contributions from the π-systems of the ppy ligands of **1**, but the acac$^-$ ligand did not contribute very much to the orbitals in **2**. Notably, the other difference, presumably associated with this structural change between **1** and **2**, is that the iridium d_{z^2} orbital is the dominant metal orbital in the HOMO of **2**. Based on the comparison

Table 1 Photophysical and thermal properties for the new metallophosphors **1** and **2**

Complexes	Absorption $\lambda_{abs}{}^a$ (nm)	Emission λ_{em} (nm)	$\Phi_P{}^b$	$\tau_p{}^c$ (μs)	$\tau_r{}^d$ (μs)	$\Delta T_{5\%}/T_g{}^e$ (°C)
1	245 (4.8), 291 (6.2), 364 (1.3)	495	0.70	2.0	2.86	403/132
2	254 (3.1), 281 (3.7), 364 (0.3)	502	0.73	3.0	4.11	356/116

[a] Measured in CH_2Cl_2 at 293 K at a concentration of 10^{-5} M and log ε values are shown in parentheses. [b] In CH_2Cl_2 at 293 K relative to *fac*-Ir(ppy)$_3$ ($\Phi_P = 0.40$), $\lambda_{ex} = 400$ nm. [c] The phosphorescent lifetimes were measured in freeze–pump–thaw degassed CH_2Cl_2 solutions at 293 K at a sample concentration of *ca.* 10^{-5} M and the excitation wavelength was set at 400 nm. [d] The triplet radiative lifetimes were deduced from $\tau_r = \tau_p/\Phi_P$. [e] $\Delta T_{5\%}$ is the 5% weight-reduction temperature and T_g is the glass-transition temperature.

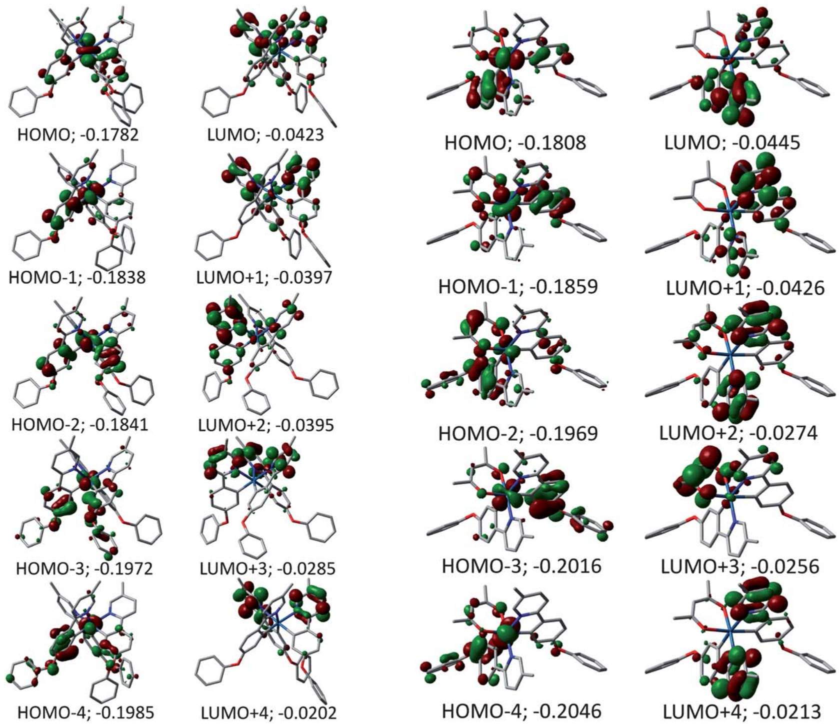


Fig. 4 Representations of the frontier MOs for an optimized geometry of the Ir^{III} complexes **1** (left) and **2** (right).

between the computed HOMO and LUMO electronic density data (Table S3†), the large decrease in metal electronic density on going from the HOMO to the LUMO confirms the strong metal-to-ligand character for this transition. This comment is relevant to the TD-DFT computations presented below as the lowest energy transition is almost exclusively composed of this HOMO → LUMO transition for both complexes. By examining Table S3† and Fig. 5, the HOMO → LUMO transition for **2** is best described as metal/ligand-to-ligand charge-transfer (MLL′CT).

TD-DFT computations were performed to confirm the nature of the lowest energy excited states. Table S4† provides the calculated transition energies, position of the 0–0 peaks, oscillator strengths (*f*) and major contributions for the 20 first transitions for **1** and **2**. The first calculated transitions are placed at 416.8 and 421.5 nm for **1** and **2**, respectively. These values fall perfectly within the first absorptions of these compounds which appear as low-intensity shoulders on the red side of the absorption. Interestingly, the lowest energy transition is composed almost exclusively of HOMO → LUMO, hence reinforcing the assignment proposed from the DFT computations described above and in agreement with the literature findings.[17] The first 100 electronic transitions were computed (see Fig. S1 and S2†) and a spectrum is generated by plotting the

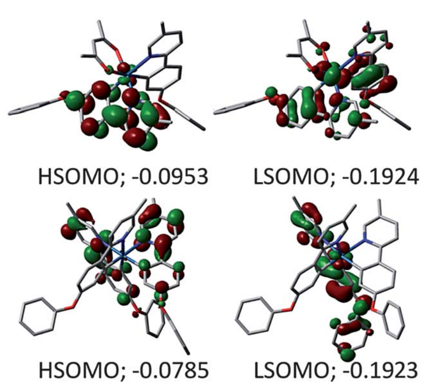


Fig. 5 Representations of the HSOMO and LSOMO for **2** (up) and **1** (below).

oscillator strength *versus* the position of the pure electronic transitions. By assigning a thickness of 1000 cm^{-1}, spectra as shown in Fig. 6 are generated. These calculated spectra do not include the presence of vibronic progressions but the shape of the spectra compares very favorably to the experimental ones (see Fig. 3). Because of the similarity between the HSOMO and LUMO and the LSOMO and HOMO (shown in Fig. 5), the contribution to the first spin-forbidden singlet-triplet transition is reasonably assumed to be composed of the same transition (*i.e.* HOMO → LUMO). Hence, these computations support the assignment of the nature of the lowest energy excited states as described above.

The lowest energy triplet excited state has also been addressed by optimizing the geometry in this state. Then the calculation of the total energy is compared to that of the ground state (S_0). The difference between both T_1 and S_0 should allow the estimation of the position of the calculated 0–0 peaks of the phosphorescence (Table S5†). These are computed to be around 447 and 495 nm for **1** and **2**. These values fall indeed in the blue region of the phosphorescence spectra. The lowest and highest energy semi-occupied MOs (LSOMO and HSOMO, respectively) for **1** and **2** bear important similarity with the HOMO and LUMO of the latter, meaning that the nature of these triplet excited states remains essentially the same (*i.e.* MLCT and M/LLCT, respectively).

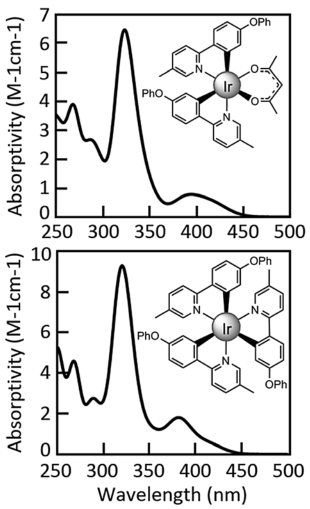


Fig. 6 Calculated spectra of **2** (up) and **1** (bottom).

Electrochemical characterization

The electrochemical behaviors of these Ir^{III} complexes were investigated by cyclic voltammetry (CV) using ferrocene as the internal standard in THF solutions of the complexes containing 0.1 M tetrabutylammonium hexafluorophosphate ($[Bu_4N]PF_6$) as the supporting electrolyte. Table 2 presents the data of the energy levels for **1** and **2** obtained both from CV measurements and TD-DFT calculations. For comparison purposes, the relevant data of $Ir(ppy)_3$ and $Ir(ppy)_2(acac)$ are also provided in Table 2.[18] Complexes **1** and **2** showed very similar oxidation behavior with an oxidation couple between 0.3 V and 0.4 V, which can be safely assigned to the Ir^{IV}/Ir^{III} oxidation. Cathodic sweeps showed an irreversible wave at −2.90 V and −2.89 V for **1** and **2**. The ancillary $acac^-$ ligand also stabilizes the HOMO level of the heteroleptic Ir^{III} complex **2**, because of its stronger ligand-field strength. By comparing with the parent complexes *fac*-$Ir(ppy)_3$ and $Ir(ppy)_2(acac)$, the introduction of the phenoxy group on the phenyl ring of the ppy sketch increases the energy gap (E_g) of **1** and **2**. However, owing to the compensation from the electron-donating character of the methyl substituent, the enlargements of E_g are slight with respect to those of the parent molecules. Both the experimental CV data and theoretical calculations confirm this change (see Table 2).

Electrophosphorescent monochromatic OLED characterization

To investigate the electroluminescent performances of **1** and **2**, a series of monochromatic OLEDs were designed by the vacuum evaporation method because of the good thermal stability of the phosphors. The configuration we adopted for the OLEDs is ITO/NPB (40 nm)/Ir^{III} (x wt%):CBP (20 nm)/BCP (10 nm)/Alq_3 (30 nm)/LiF (1 nm)/Al (100 nm). Fig. 7 depicts the structure of the multi-layered OLEDs and the compounds involved in the fabrication of these devices. Indium tin oxide (ITO) was chosen as the anode of the devices due to its matched work function (4.5–4.8 eV), good conductivity, high transparency and excellent stability.[23] 4,4′-Bis[*N*-(1-naphthyl)-*N*-phenylamino]biphenyl (NPB) acts as a hole-transporting (HT) layer, 4,4′-*N*,*N*′-dicarbazolebiphenyl (CBP) as a small-molecule host material for the electrophosphors **1** and **2**, 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP) as a hole-blocker, tris(8-hydroxyquinolinato)aluminum (Alq_3) as an electron-transporter, and LiF as an electron-injection (EI) layer. Low

Table 2 Comparison of electronic properties of the new phenoxy-containing Ir^{III} complexes and their parent complexes *fac*-$Ir(ppy)_3$ and $Ir(ppy)_2(acac)$

Compound	$E_{1/2}^{ox\,a}$ (V)	$E_{1/2}^{red\,a}$ (V)	HOMO (eV) Experimental^b/theoretical	LUMO (eV) Experimental^c/theoretical	$E_g{}^d$ (eV)
1	0.32	-2.90^e	−5.12/−4.85	−1.90/−1.15	3.22/3.70
2	0.39	-2.89^e	−5.19/−4.92	−1.91/−1.21	3.28/3.71
$Ir(ppy)_3{}^f$	0.26	−2.77	−5.06/−4.85	−2.03/−1.21	3.03/3.66
$Ir(ppy)_2(acac)^g$	^{h}	^{h}	−5.15/−5.08	−2.34/−1.57	2.81/3.51

^{a} 0.1 M $[Bu_4N]PF_6$ in THF, *versus* Fc/Fc^+ couple. ^{b} HOMO = $-(4.8\text{ eV} + E_{1/2}^{ox})$. ^{c} LUMO = $-(4.8\text{ eV} + E_{1/2}^{red})$. ^{d} E_g = LUMO − HOMO. ^{e} Irreversible reduction wave. ^{f} Experimental data are taken from ref. 19 and theoretical data are taken from ref. 20. ^{g} Experimental data are taken from ref. 21 and theoretical data are taken from ref. 22. ^{h} Not reported.

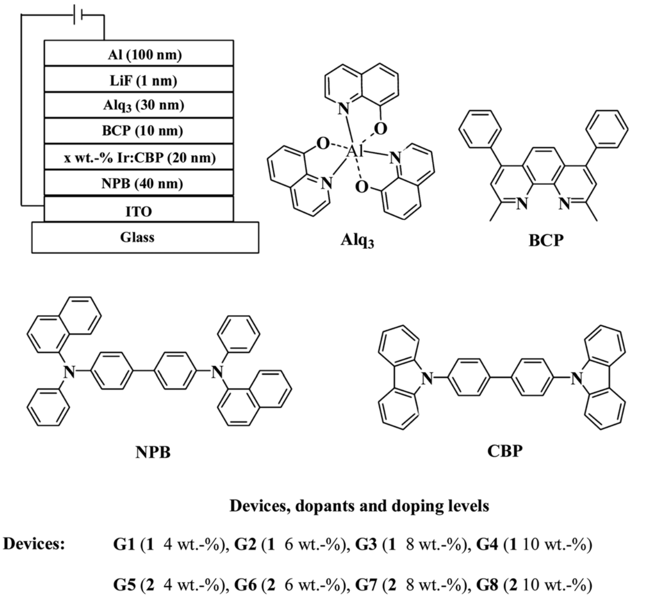


Fig. 7 The general configuration for the optimized electrophosphorescent OLED devices and molecular structures of the relevant compounds used in these devices.

work function aluminum was used as the cathode. To optimize the device efficiency, concentration dependence experiments were carried out in the range of 4 to 10 wt% for **1** and **2** to give green-emitting OLEDs labeled as **G1** to **G4** and **G5** to **G8**, respectively.

All of the devices emit intense green electroluminescence (EL) and the emission peaks for **G1** to **G4** and **G5** to **G8** are located at 496 nm and 504 nm, respectively. In each case, the EL spectrum exhibits the same emission pattern as the PL spectrum, which indicates that the EL originates from the same triplet state of the Ir^{III} cyclometallates (see Fig. 8). No emission from CBP or Alq_3 was observed in each device, suggesting an efficient energy transfer from the host to the Ir^{III} complex *via* the Dexter mechanism, as well as the effective hole-blocking effect of BCP.[2a] The performance data of **G1** to **G8** are listed in Table 3. Device **G4** doped with 10 wt% **1** gave very stable EL spectra under different voltages (see Fig. S3†). As indicated from the *J*–*V*–*L* curves in Fig. 9, device **G4** exhibited the best performance for the homoleptic species, with a maximum brightness (L_{max}) of 46 404 cd m^{-2} at 15.1 V, a peak external quantum efficiency (η_{ext}) of 10.0%, a luminance efficiency (η_L) of 31.1 cd A^{-1} and a power efficiency (η_P) of 14.5 lm W^{-1}. The optimized device **G7** based on the heteroleptic complex **2** exhibited a very impressive EL performance with the highest η_{ext} of 11.7%, η_L of 38.1 cd A^{-1} and η_P of 31.8 lm W^{-1} (Fig. 10).

Motivated by the good performance of **1** and **2** in the typical OLEDs, we pursued optimizing the configuration of OLEDs. Firstly, the devices have been electrically optimized by replacing the electron-transporting (ET) material Alq_3 and hole-blocking (HB) material BCP with a single layer of TPBi (1,3,5-tris[*N*-(phenyl)benzimidazole]benzene) (Fig. 11). Since TPBi has significantly higher electron mobility than that of BCP, the electron transport from TPBi to the emissive layer is more effective than that of BCP, which would help further enhance the efficiencies of OLEDs.[24] Secondly, the devices have been optically optimized by increasing the thickness of the HT layer (NPB) from 40 nm to 60 nm so that the green emission can be remarkably enhanced due to the constructive interference in the cavity.[25] With the concentration of **1** or **2** varied from 4 wt% to 12 wt%, green-emitting devices **G9** to **G14** were fabricated by thermal sublimation (Fig. 11).

As shown in Table 3, all turn-on voltages ($V_{turn\text{-}on}$) of the devices are below 3.8 V. According to the CIE coordinates of the EL spectra, all of our devices emit light in the green region. By using TPBi with high electron mobility and by thickening the HT layer, devices **G9** to **G14** showed higher efficiencies than the BCP-based devices we investigated above (**G1** to **G8**). In all cases, as the voltage increased, the luminance of devices rose dramatically and the maximum luminance was obtained at the highest driving voltage. Furthermore, the maximum luminance increases as the dopant concentration increases. For comparison purpose, we also fabricated the benchmark devices **R1** to **R3** containing 4, 8 and 12 wt% *fac*-$Ir(ppy)_3$ with the same structure as devices **G9** to **G14** (Fig. 12).

As indicated in Table 3, the best doping level for **1** in this series of OLEDs is 12 wt% and device **G11** exhibited high efficiency with maximum η_{ext} of 22.5%, η_L of 76.2 cd A^{-1} and η_P of 72.8 lm W^{-1}. The best doping level for **2** is 8 wt% because **G13**

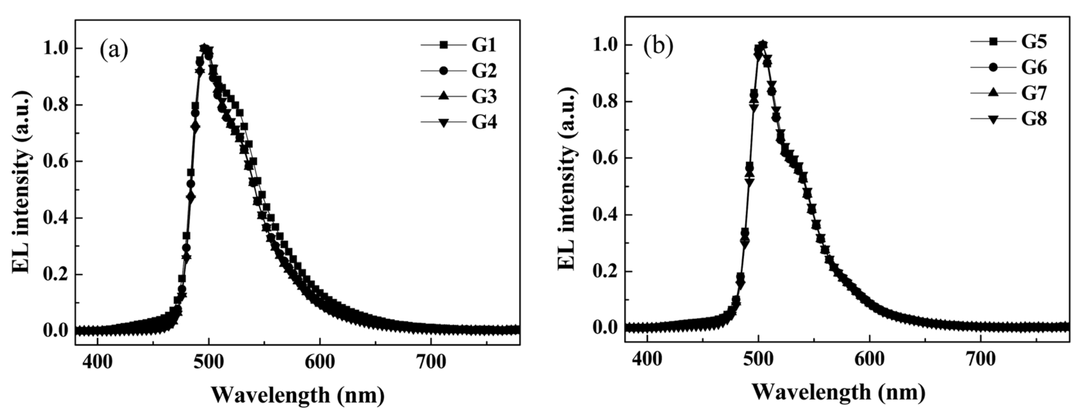


Fig. 8 The EL spectra for (a) devices **G1** to **G4** and (b) devices **G5** to **G8** under the driving voltage of 8 V.

Table 3 The EL performance of the OLEDs based on various Ir^{III} cyclometallates

	Dopant	$V_{turn\text{-}on}$ (V)	Luminance L^a (cd m^{-2})	$\eta_{ext}{}^a$ (%)	$\eta_L{}^a$ (cd A^{-1})	$\eta_P{}^a$ (lm W^{-1})	$\lambda_{max}{}^b$ (nm)
G1	**1** (4 wt%)	3.7	15 637 (14.5)	6.0 (6.7)	18.3 (6.5)	9.7 (5.9)	496 (0.24, 0.56)
G2	**1** (6 wt%)	3.5	30 436 (16.3)	8.8 (3.9)	26.8 (3.9)	21.6 (3.7)	496 (0.23, 0.57)
G3	**1** (8 wt%)	3.1	38 834 (15.1)	9.4 (5.9)	29.0 (5.9)	23.9 (3.9)	496 (0.23, 0.59)
G4	**1** (10 wt%)	3.1	46 404 (15.1)	10.0 (7.3)	31.1 (7.3)	14.5 (6.7)	496 (0.23, 0.59)
G5	**2** (4 wt%)	3.7	36 261 (14.7)	8.5 (4.5)	27.2 (4.5)	19.0 (4.5)	504 (0.23, 0.60)
G6	**2** (6 wt%)	3.5	41 605 (14.5)	11.0 (4.1)	35.3 (4.1)	28.7 (4.9)	504 (0.23, 0.60)
G7	**2** (8 wt%)	3.5	43 203 (14.9)	11.7 (4.3)	38.1 (4.3)	31.8 (3.5)	504 (0.23, 0.61)
G8	**2** (10 wt%)	3.5	47 294 (14.3)	9.2 (7.7)	30.2 (7.7)	12.4 (6.9)	504 (0.23, 0.62)
G9	**1** (4 wt%)	3.8	41 174 (15)	18.3 (3.8)	61.8 (3.8)	51.1 (3.8)	500 (0.29, 0.59)
G10	**1** (8 wt%)	3.4	47 515 (15)	21.1 (3.4)	72.3 (3.4)	66.8 (3.4)	500 (0.29, 0.59)
G11	**1** (12 wt%)	3.2	74 133 (15)	22.5 (3.4)	76.2 (3.6)	72.8 (3.2)	500 (0.29, 0.59)
G12	**2** (4 wt%)	3.6	75 030 (15)	22.8 (4.2)	78.8 (4.2)	62.7 (3.6)	504 (0.25, 0.61)
G13	**2** (8 wt%)	3.2	102 525 (15)	24.5 (3.8)	84.6 (3.8)	77.6 (3.2)	504 (0.26, 0.63)
G14	**2** (12 wt%)	3.0	131 497 (15)	24.3 (3.8)	83.8 (3.8)	76.1 (3.0)	504 (0.26, 0.63)
R1	$Ir(ppy)_3$ (4 wt%)	3.4	62 829 (15)	15.7 (3.8)	56.8 (3.8)	46.9 (3.6)	516 (0.32, 0.62)
R2	$Ir(ppy)_3$ (8 wt%)	3.2	56 572 (15)	11.3 (3.6)	40.7 (3.6)	38.7 (3.2)	516 (0.32, 0.62)
R3	$Ir(ppy)_3$ (12 wt%)	3.0	94 590 (15)	9.7 (4.6)	35.1 (4.6)	27.2 (3.8)	516 (0.32, 0.62)

[a] Maximum values of the devices. Values in parentheses are the voltages at which they were obtained. [b] Values were collected at 8 V and CIE coordinates (x, y) are shown in parentheses.

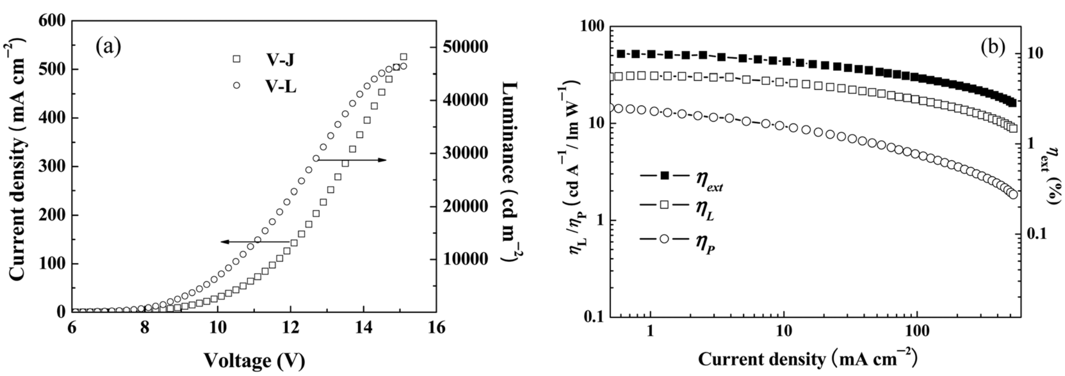


Fig. 9 (a) The current density (*J*)–voltage (*V*)–luminance (*L*) curves and (b) the relationship between EL efficiencies and current density for the device **G4**.

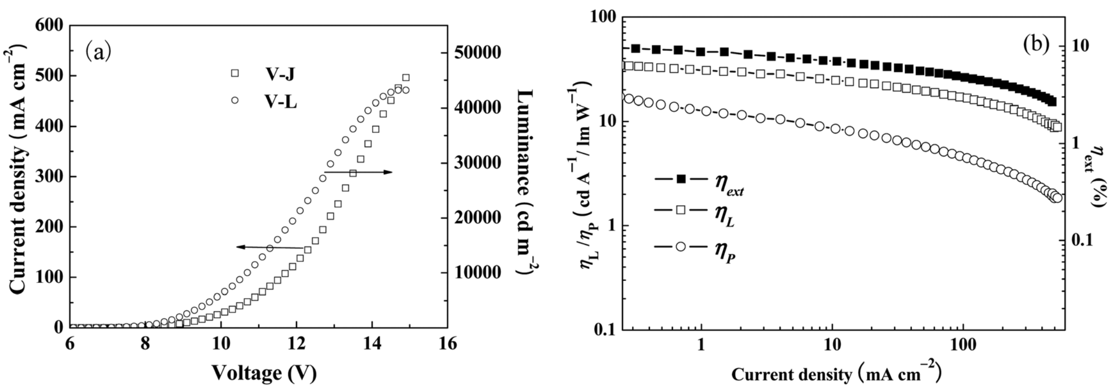


Fig. 10 (a) The *J*–*V*–*L* curves and (b) the relationship between EL efficiencies and current density for the device **G7**.

gave the best performance. **G13** showed higher current density than other devices (Fig. 12a). Device **G13** is the brightest with a maximum luminance of 102 525 cd m^{-2} (Fig. 12b) and showed η_{ext} as high as 24.5% at 0.05 mA cm^{-2}. Owing to the triplet–triplet annihilation, at J = 1, 10 and 50 mA cm^{-2}, the η_{ext} decreases from the peak value of 24.5% to 19.9%, 14.4% and 13.6%, respectively (Fig. 12e).[26] Furthermore, **G13** exhibited superior efficiency values with the peak η_L of 84.6 cd A^{-1} (Fig. 12c) and η_P of 77.6 lm W^{-1} (Fig. 12d). When doped with 12 wt% of **2**, device **G14** showed a great peak luminance of

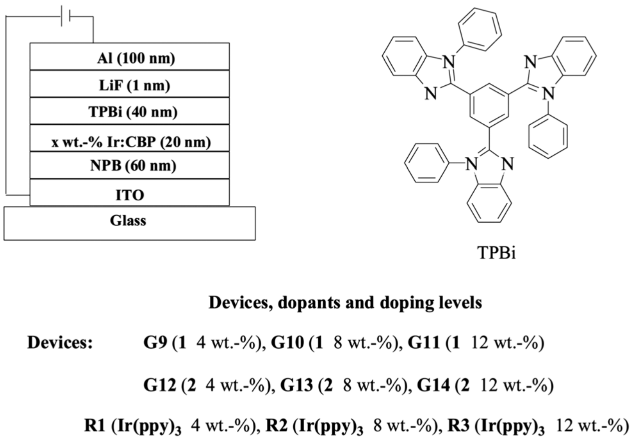


Fig. 11 The optimized configuration of OLED devices and the molecular structure of TPBi.

131 497 cd m^{-2} at the maximal driving voltage of 15 V. For the devices doped with the benchmark green phosphor Ir(ppy)$_3$, **R1** doped with 4 wt% *fac*-Ir(ppy)$_3$ only produced maximum values of η_L, η_P and η_{ext} at 56.8 cd A^{-1}, 46.9 lm W^{-1} and 15.7%, respectively. The *fac*-Ir(ppy)$_3$ doped devices suffered from a serious self-quenching effect, since the efficiencies decreased dramatically as the concentration of *fac*-Ir(ppy)$_3$ increased from 4 wt% to 12 wt%. For instance, **R3** doped with 12 wt% *fac*-Ir(ppy)$_3$ only led to the peak efficiencies at η_{ext} of 9.7%, η_L of 35.1 cd A^{-1} and η_P of 27.2 lm W^{-1}, which are almost half of the values of **R1**. However, such a situation did not severely occur with the **1** and **2** doped OLEDs. One possible explanation is that through the bifunctionalization with a phenoxy group and a methyl substituent on ppy, the intermolecular distance increased, thereby decreasing the self-quenching effect between the host material and metal phosphor. Such high efficiencies achieved by our devices indicate that by modifying the ppy ligand in this way, the

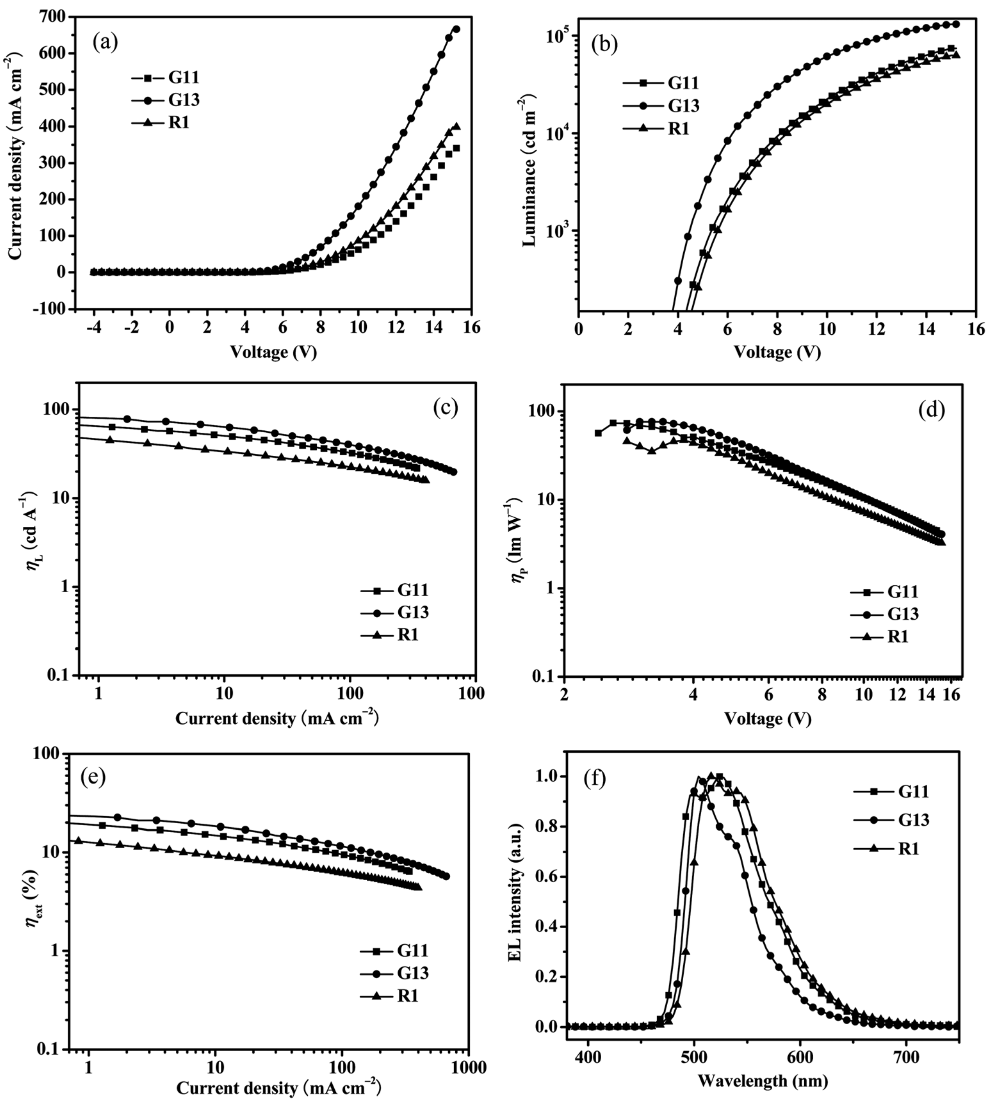


Fig. 12 Charts for the three best devices **G11**, **G13** and **R1** doped with **1**, **2** and *fac*-Ir(ppy)$_3$, respectively. (a) *J*–*V* curves; (b) *L*–*V* curves; (c) η_L–*J* curves; (d) η_P–*V* curves; (e) η_{ext}–*J* curves and (f) EL spectra.

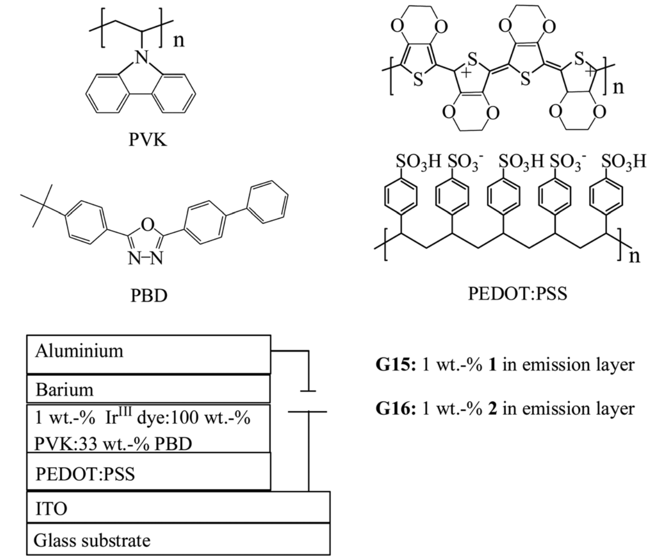


Fig. 13 The general configuration for the solution-processed PLED devices and molecular structures of the relevant compounds used in these devices.

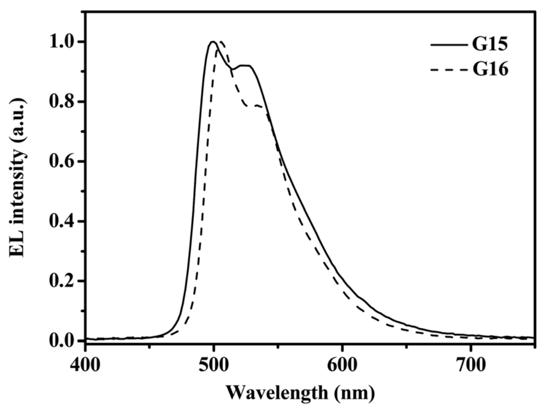


Fig. 14 EL spectra for PLEDs **G15** and **G16**.

luminescent feature of our Ir^{III} cyclometallates has improved, leading to outstanding OLED performances.

The excellent performances of **1** and **2** in evaporated OLEDs prompted us to explore their application in solution-processed polymer light-emitting devices (PLEDs) *via* the convenient spin-coating method. The device configuration of **G15** and **G16** is shown in Fig. 13. Poly(3,4-ethylenedioxy-thiophene):poly(styrene sulfonate) (PEDOT:PSS) was coated on the ITO anode. A poly(*N*-vinylcarbazole) (PVK) host system was adopted for the emission layer. Semi-conducting PVK is an efficient HT material and an ET 2-(4-biphenylyl)-5-(4-*tert*-butylphenyl)-1,3,4-oxa-diazole (PBD) layer was used for balancing charge transport within this system.[27] The emission layer of the PLED **G15** was comprised of the composition of 1 wt% **1** : 100 wt% PVK : 33 wt% PBD while **G16** consisted of 1 wt% **2** : 100 wt% PVK : 33 wt% PBD. Finally, the devices were capped with the aluminum/barium alloy bilayer cathode.

The EL spectra (Fig. 14) of devices **G15** and **G16** are almost the same as the PL spectra of **1** and **2** in CH_2Cl_2 under room temperature. As shown in Table 4, after doping with 1 wt% **1**, device **G15** lighted up under a driving voltage of 6.6 V, which gave the peak η_{ext} of 14.4%, η_L of 39.5 cd A^{-1} and η_P of 12.4 lm W^{-1}. Device **G16** with 1 wt% **2** showed a low $V_{turn-on}$ of 4.0 V and respectable EL results with η_{ext} of 12.6%, η_L of 29.6 cd A^{-1} and η_P of 18.1 lm W^{-1}. Both PLEDs **G15** and **G16** emitted strong green light with luminance above 10 000 cd m^{-2} at 15 V (Fig. S5†). However, it is not clear at this stage why PLEDs based on **1** are more efficient than PLEDs based on **2**. We speculate that it can be attributed to a more balanced charge transport in PLEDs based on **1**. Work is currently underway to understand this issue.

Three-element WOLED characterization

White organic light-emitting diode (WOLED) technology is a major focus of OLED research owing to its potential application for solid-state lighting sources and backlights in flat displays.[28] It is known that white light is generated by additively mixing red (R), green (G) and blue (B) emission with equal intensity. So, most white-emitting devices consist of three light-emitting layers with each layer doped by a luminescent dopant emitting one of the primary colors. Since **2** exhibited better performances than **1** in evaporated OLED devices, we used **2** as a green phosphor in the design of three-component WOLEDs. A series of hybrid WOLEDs with the structure: ITO/NPB (40 nm)/6 wt% Ir(2-phq)$_2$(acac):CBP (7 nm)/12 wt% **2**:CBP (x nm)/CBP (3 nm)/10 wt% BCzVBi:CBP (5 nm)/TPBi (40 nm)/LiF (1 nm)/Al (100 nm) (BCzVBi = 4,4′-bis(9-ethyl-3-carbazovinylene)-1,1′-biphenyl; Ir(2-phq)$_2$(acac) = bis(2-phenylquinoline)(acetylacetonate)iridium(III)) were built (Fig. S6† and Table 5). The thickness of the emission layer doped with **2** was varied from 1 to 2 to 3 nm to tune the emission intensity of the green light accurately, and the devices were abbreviated as **W1**, **W2** and **W3**. It is noted that the emission peak of **2** is located at 504 nm, which is too close to the emission peak of 475 nm for the blue phosphorescent emitter FIrPic bis(3,5-difluoro-2-(2-pyridyl)phenyl)(2-carboxypyridyl)iridium(III), but

Table 4 The EL data of the PLEDs doped with **1** and **2**

	Dopant	$V_{turn-on}$ (V)	Luminance L^a (cd m^{-2})	$\eta_{ext}{}^a$ (%)	$\eta_L{}^a$ (cd A^{-1})	$\eta_P{}^a$ (lm W^{-1})	$\lambda_{max}{}^b$ (nm)
G15	**1** (1 wt%)	6.6	14 333 (17.0)	14.4 (10.0)	39.5 (10.0)	12.4 (10.0)	499 (0.28, 0.59)
G16	**2** (1 wt%)	4.0	10 740 (14.7)	12.6 (8.0)	29.6 (8.0)	18.1 (4.7)	505 (0.28, 0.61)

^{a} Maximum values of the devices. Values in parentheses are the voltages at which they were obtained. ^{b} CIE coordinates (x, y) are shown in parentheses.

Table 5 Key characteristics of the three-component WOLEDs with different green emitters

	$V_{turn-on}$ (V)	L_{max} (cd m^{-2}) @15 V	η_{ext}[a] (%)	η_L[a] (cd A^{-1})	η_P[a] (lm W^{-1})	CIE (x, y)[b]	CRI/CCT (K)
W1	3.8	23 940	6.8 (5.8)	12.8 (5.8)	8.8 (4.4)	0.38, 0.33	64/3517
W2	4	28 714	7.0 (6.0)	14.0 (6.0)	5.0 (8.1)	0.40, 0.37	73/3309
W3	4	34 908	7.5 (5.8)	15.3 (5.8)	9.1 (4.8)	0.38, 0.37	76/4038

[a] Maximum values of the devices. Values in parentheses are the voltages at which they were obtained. [b] CIE coordinates (x, y) at 8 V.

is also too far away from the emission peak of 616 nm for the conventional red phosphorescent emitter $Ir(btp)_2(acac)$ (bis(2-benzo[*b*]thiophen-2-yl-pyridine)(acetylacetonate)iridium(III)). Hence, both FIrPic and $Ir(btp)_2(acac)$ are not good partners for **2** when they are used to construct WOLEDs, as the resultant white spectra would exhibit low CRI due to the unevenly separated R, G and B peaks.[29] To realize a balanced white emission with high CRI and evenly separated R, G and B peaks, a deep blue fluorescent emitter BCzVBi and an orange emitter $Ir(2\text{-}phq)_2(acac)$ were chosen.[30]

For comparison purposes, hybrid WOLEDs using the benchmark green emitter *fac*-$Ir(ppy)_3$ with an identical device structure were made. Based on the investigation of monochromatic OLEDs doped with *fac*-$Ir(ppy)_3$, the best dopant percentage was fixed at 6 wt%. Moreover, the thickness of the *fac*-$Ir(ppy)_3$ doped emission layer was also carefully tuned from

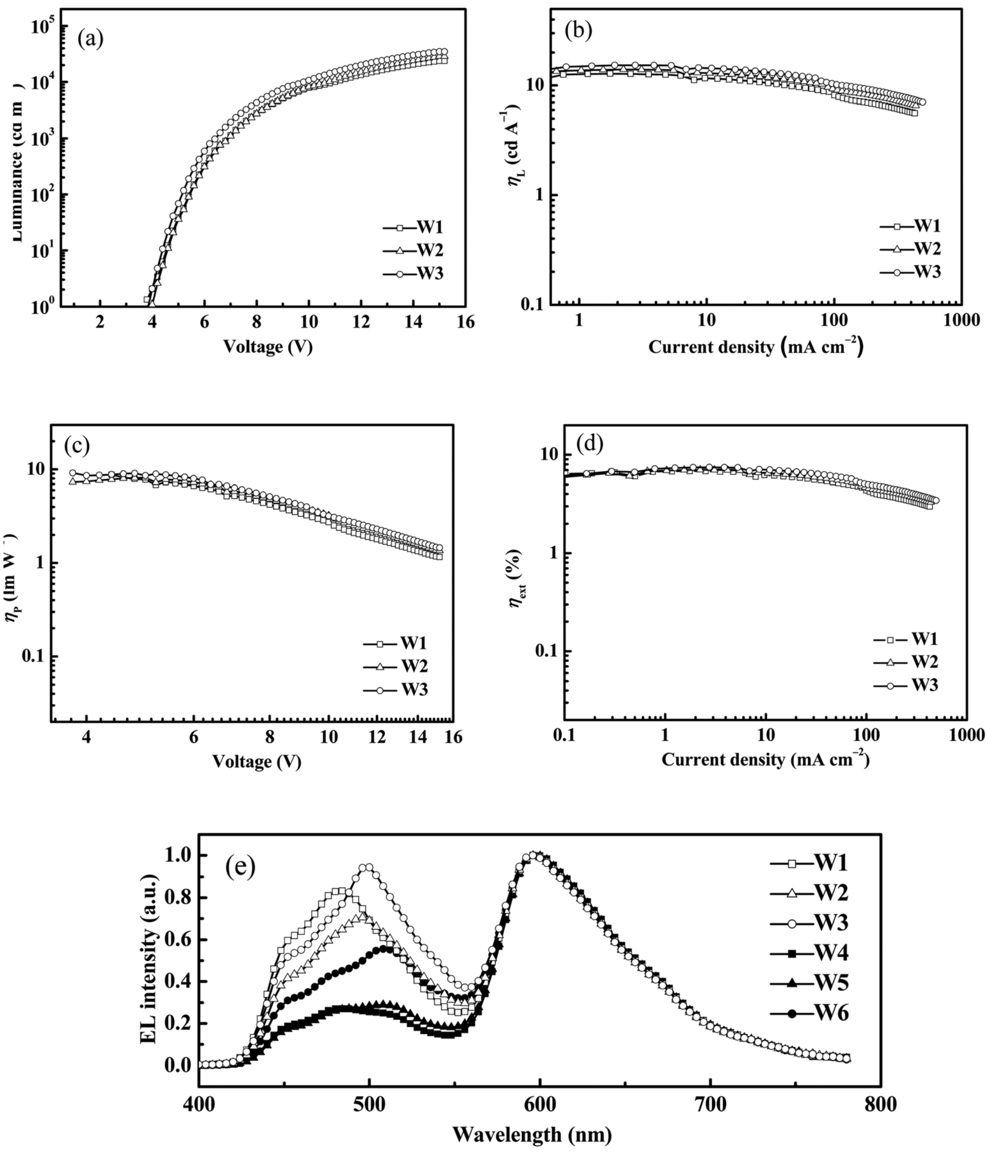


Fig. 15 Charts for the hybrid WOLEDs **W1–W3**. (a) *L–V* curves; (b) η_L–*J* curves; (c) η_P–*V* curves; (d) η_{ext}–*J* curves and (e) EL spectra at 8 V (the spectra of **W4–W6** are also included).

1 to 2 to 3 nm (similar to the **2**-doped WOLEDs) to provide reference WOLEDs labeled as **W4**, **W5** and **W6**. The performance characteristics of **W4–W6** are listed in ESI.†

Fig. 15a shows the typical *L–V* characteristics of the devices **W1–W3**. The luminance rose with an increase of the driving voltage. All of the devices turned on at or below 4 V for 1 cd m^{-2} and the devices with a thicker **2**-doped emission layer exhibited higher luminance. For example, at a driving voltage of 15 V, the luminance increases from 23 940 cd m^{-2} for **W1** (1 nm) to 28 714 cd m^{-2} and 34 908 cd m^{-2} for **W2** (2 nm) and **W3** (3 nm), respectively, mainly due to the high efficiency of the green emission. To better illustrate the performance of these WOLEDs, the efficiency *versus* driving current density or voltage curve is plotted (Fig. 15b–d). It was shown that the devices with **2** showed comparable EL efficiencies with that of *fac*-Ir(ppy)$_3$ (see ESI†) but their white color Commission Internationale de L'Eclairage (CIE) coordinates are improved relative to Ir(ppy)$_3$. As shown in Fig. 15e, the EL spectra of WOLEDs exhibited two major emission bands. From 420 to 550 nm, the broad emission is composed of a fluorescent blue light generated by BCzVBi and an intense green light attributed to **2** or *fac*-Ir(ppy)$_3$. Another low-lying emission band from 550 to 780 nm is caused by the red phosphor Ir(2-phq)$_2$(acac). The CIE coordinates, color rendering index (CRI) and correlated color temperature (CCT) of **W1–W3** are listed in Table 5. Remarkably, the CIE coordinates of the WOLEDs doped with **2** are located nearer to the equal-energy-white-point (0.33, 0.33) than those with *fac*-Ir(ppy)$_3$ (Fig. 16). For instance, the CIE coordinates of **W1** are (0.38, 0.33) while those of **W6** are (0.43, 0.39). After amplifying the intensity of the green emission generated from **2**, the device **W3** gave the best CRI value as high as 76 and the CCT of 4038 K, which is close to the direct sunlight at noon (CCT of 4874 K). The emission color of WOLEDs doped with *fac*-Ir(ppy)$_3$ is only near the yellow–orange region, because the green emission is relatively weaker than the red phosphorescence.

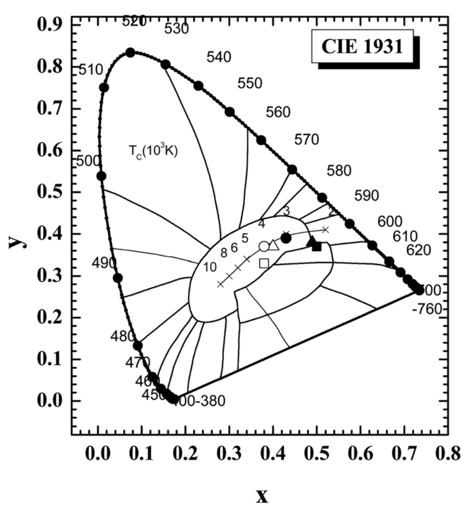


Fig. 16 The CIE 1931 chromaticity diagram showing the color coordinates of the resulting WOLEDs: **W1** (□); **W2** (△); **W3** (○); **W4** (■); **W5** (▲); **W6** (●).

Conclusion

Two new functionalized Ir^{III} triplet emitters with a flexible phenoxy group and a methyl substituent on ppy (**1** and **2**) were synthesized and fully characterized. Owing to the excellent electronic character associated with such a functional ligand, these favorable features were shown to significantly improve the EL performance of the OLEDs. Similarly, the monochromatic green OLEDs exhibited excellent EL abilities with peak η_{ext} of 22.5%, η_L of 76.2 cd A^{-1} and η_P of 72.8 lm W^{-1} for **1** and peak η_{ext} of 24.5%, η_L of 84.6 cd A^{-1} and η_P of 77.6 lm W^{-1} for **2**. All of them outperformed the devices doped with the state-of-the-art green emitter *fac*-Ir(ppy)$_3$ and Ir(ppy)$_2$(acac) under the same device configuration. For the three-color WOLEDs, **2** also showed comparable efficiencies to and better CIE values than *fac*-Ir(ppy)$_3$. All the experimental data confirmed that the green triplet emitters **1** and **2** are outstanding candidates in the applications of electrophosphorescent OLEDs and WOLEDs. These excellent device performances clearly reveal the great potential of this class of green metallophosphors in both thermally evaporated small-molecule OLEDs and solution-processed PLEDs.

Experimental

General information

All reactions were carried out under nitrogen atmosphere. Solvents were carefully dried and distilled from appropriate drying agents prior to use. Commercially available reagents were used without further purification unless otherwise stated. All reactions were monitored by thin-layer chromatography (TLC) with Merck pre-coated aluminum plates. Flash column chromatography and preparative TLC were carried out using silica gel from Merck (230–400 mesh). Fast atom bombardment (FAB) mass spectra were recorded on a Finnigan MAT SSQ710 system. Proton and $^{13}C\{^1H\}$ NMR spectra were measured in $CDCl_3$ on a Bruker Ultra-shield 400 MHz spectrometer and chemical shifts are quoted relative to tetramethylsilane.

Physical measurements

UV/Vis spectra were obtained on a Hewlett Packard 8453 spectrophotometer. The photoluminescent properties of the compounds were examined using a PTI Time Master Model C-720 spectrophotometer. The phosphorescence properties were determined in CH_2Cl_2 solutions at 293 K against *fac*-[Ir(ppy)$_3$] standard ($\Phi_P = 0.40$).[31] For lifetime measurements, the 337 nm line of a N_2 laser was used as the excitation light source. The data analysis was conducted by iterative convolution of the luminescence decay profile with the instrument response function using the software package provided by PTI Instruments. Electrochemical measurements were made using a Princeton Applied Research model 2273A potentiostat at a scan rate of 100 mV s^{-1}. A conventional three-electrode configuration consisting of a glassy carbon working electrode, a Pt-wire counter electrode, and a Ag/Ag$^+$ reference electrode was used. The supporting electrolyte was 0.1 M $[Bu_4N]PF_6$ in THF. Ferrocene was added as a calibrant after each set of measurements, and all potentials reported are quoted with reference to the ferrocene–ferrocenium (Fc/Fc$^+$) couple. The oxidation (E_{ox}) and reduction (E_{red}) potentials were used to determine the HOMO and LUMO energy levels using the equations $E_{HOMO} = -(E_{ox} + 4.8)$ eV and $E_{LUMO} = -(E_{red} + 4.8)$ eV which were calculated using the internal standard ferrocene value of −4.8 eV with respect to the vacuum level.[32] Differential scanning calorimetry (DSC) was performed on a Perkin Elmer Pyris Diamond DSC unit under a nitrogen flow at a heating rate of 20 °C min^{-1} to obtain the glass transition temperature (T_g). The thermal gravimetric analysis (TGA) was conducted on a Perkin Elmer TG-6 instrument under nitrogen at the heating rate of 5 °C min^{-1}.

Synthesis of L

$Pd(PPh_3)_4$ (200 mg) was added to a mixture of 4-phenoxyphenylboronic acid (1.70 g, 7.94 mmol) and 2-chloro-5-methylpyridine (0.405 g, 3.18 mmol) in toluene (100 mL) and 2 M aqueous Na_2CO_3 (8 mL) under an inert atmosphere of nitrogen. The reaction was heated to 110 °C for 48 h. After being cooled to room temperature, water was added and the solution mixture was extracted with ethyl acetate. The combined organic layer was dried by Na_2SO_4 and was filtered and concentrated under reduced pressure. The residue was purified through column chromatography over silica gel eluting with hexane–ethyl acetate (5 : 1, v/v) to give the ligand **L** as a white powder in 63% yield. Spectral data: MS (FAB): *m*/*z* 262 $[M + 1]^+$. ^{1}H NMR ($CDCl_3$): δ (ppm) 8.45 (s, 1H, Ar), 7.93–7.91 (m, 2H, Ar), 7.50–7.48 (d, J = 8.0 Hz, 1H, Ar), 7.43–7.40 (m, 1H, Ar), 7.32–7.28 (m, 2H, Ar), 7.09–7.01 (m, 5H, Ar), 2.27 (s, 3H, CH_3); ^{13}C NMR ($CDCl_3$): δ (ppm) 157.94, 157.02, 154.12, 150.03, 137.36, 134.52, 131.27, 129.88, 128.22, 123.53, 119.60, 119.13, 118.89 (Ar), 18.17 (CH_3). Anal. calcd for $C_{18}H_{15}NO$: C, 82.32; H, 5.79; N, 5.36. Found: C, 82.12; H, 5.98; N, 5.55%.

Synthesis of 1

Under a N_2 atmosphere, cyclometallating ligand **L** (0.51 g, 1.94 mmol) and $[Ir(acac)_3]$ (0.23 g, 0.48 mmol) were heated to 220 °C in glycerol for 20 h. The reaction mixture was cooled to room temperature and water was added. After extraction with CH_2Cl_2 (3 × 60 mL), the organic phase was dried over Na_2SO_4. The solvent was removed under reduced pressure. The residue was obtained as a crude product, which was purified by chromatography on a silica column using CH_2Cl_2–hexane (4 : 1, v/v) eluent to produce **1** as a yellow solid (126 mg, 27%). Spectral data: MS (FAB): *m*/*z* 973.0 $[M + 1]^+$. ^{1}H NMR ($CDCl_3$): δ (ppm) 7.69–7.67 (d, J = 8.5 Hz, 3H, Ar), 7.52–7.49 (d, J = 8.5 Hz, 3H, Ar), 7.43–7.40 (m, 3H, Ar), 7.31 (s, 3H, Ar), 7.13–7.09 (m, 6H, Ar), 6.90–6.93 (m, 3H, Ar), 6.80–6.77 (m, 6H, Ar), 6.49–6.48 (d, J = 2.5 Hz, 3H, Ar), 6.42–6.39 (m, 3H, Ar), 2.04 (s, 9H, CH_3); ^{13}C NMR ($CDCl_3$): δ (ppm) 163.37, 162.49, 157.54, 157.47, 146.84, 139.50, 136.95, 130.70, 129.22, 126.47, 124.79, 122.09, 118.47, 118.01, 110.66 (Ar), 18.33 (CH_3). Anal. calcd for $C_{54}H_{42}N_3O_3Ir$: C, 66.65; H, 4.35; N, 4.32. Found: C, 66.45; H, 4.39; N, 4.23%.

Synthesis of 2

Under a N_2 atmosphere, ligand **L** (0.82 g, 3.16 mmol) and $IrCl_3 \cdot nH_2O$ (375 mg, 1.26 mmol) were heated to 100 °C in a mixture of 2-ethoxyethanol and water (15 mL, 3 : 1, v/v) for 16 h. The reaction mixture was cooled to room temperature and water was added. The cyclometallated Ir^{III} μ-chloro-bridged dimer was formed as a precipitate which was collected and dried under vacuum. The dimeric Ir complex, 3 equiv. of acetylacetone and 10 equiv. of Na_2CO_3 were added to 2-ethoxyethanol and the mixture was heated to 90 °C for 18 h. After cooling to room temperature and the addition of water, the brown precipitate was collected by filtration and washed with water and dried. The crude product was purified by chromatography on a silica column using CH_2Cl_2–hexane (3 : 1, v/v) eluent to produce **2** (215 mg, 21%) as a yellow solid. Spectral data: MS (FAB): *m*/*z* 811.7 $[M + 1]^+$. ^{1}H NMR ($CDCl_3$): δ (ppm) 8.17 (s, 2H, Ar), 7.47–7.45 (d, J = 8.2 Hz, 2H, Ar), 7.35–7.30 (m, 4H, Ar), 7.20–7.16 (m, 4H, Ar), 7.00–6.96 (m, 2H, Ar), 6.87–6.84 (m, 4H, Ar), 6.36–6.34 (m, 2H, Ar), 5.81–5.80 (d, J = 2.5 Hz, 2H, Ar), 5.21 (s, 1H, acac), 2.26 (s, 6H, CH_3), 1.79 (s, 6H, acac); ^{13}C NMR ($CDCl_3$): δ (ppm) 184.44 (acac), 165.02, 157.08, 156.50, 148.68, 147.48, 140.13, 137.69, 130.42, 129.25, 124.40, 122.88, 122.01, 119.58, 117.62, 110.01 (Ar), 100.54 (acac), 28.78 (acac), 18.36 (CH_3). Anal. calcd for $C_{41}H_{35}N_2O_4Ir$: C, 60.65; H, 4.34; N, 3.45. Found: C, 60.82; H, 4.45; N, 3.63%.

X-Ray crystallography

Crystals of our complexes suitable for single-crystal X-ray analyses were mounted on a glass fibre. Intensity data were collected at 293 K on a Bruker Axs SMART 1000 CCD area-detector diffractometer using graphite-monochromated Mo-Kα radiation (λ= 0.71073 Å). The collected frames were processed with the software SAINT[33] and an absorption correction was applied (SADABS)[34] to the collected reflections. The space groups of each crystal were determined from the systematic absences and Laue symmetry check and confirmed by successful refinement of the structure. The structures of all compounds were solved either by direct methods or Patterson methods (SHELXTL™)[35] and refined against F^2 by full matrix least-squares analysis. All non-hydrogen atoms were refined anisotropically. Hydrogen atoms were generated in their idealized positions and allowed to ride on their respective parent carbon atoms. The crystallographic data for the complexes (excluding structure factors) have been deposited in the Cambridge Crystallographic Data Centre with the deposition numbers CCDC-898069 (**1**) and CCDC-898070 (**2**). Crystal data for **1**: $C_{54}H_{42}IrN_3O_3$, M_w = 973.11, monoclinic, space group $P2_1/n$, a = 10.7639(6), b = 28.0012(15), c = 14.6392(8) Å, β = 90.03(1), V = 4412.3(4) Å^3, Z = 4, ρ_{calcd} = 1.465 g cm^{-3}, $\mu(Mo_{K\alpha})$ = 3.074 mm^{-1}, $F(000)$ = 1952. 27 679 reflections measured, of which 10 641 were unique (R_{int} = 0.0296). Final R_1 = 0.0285 and wR_2 = 0.0661 for 8013 observed reflections with $I > 2\sigma(I)$. Crystal data for **2**: $C_{41}H_{35}IrN_2O_4$, M_w = 811.91, monoclinic, space group $P2_1/c$, a = 10.167(1), b = 11.472(1), c = 29.660(3) Å, β = 95.018(2), V = 3446.3(7) Å^3, Z = 4, ρ_{calcd} = 1.565 g cm^{-3}, $\mu(Mo_{K\alpha})$ = 3.919 mm^{-1}, $F(000)$ = 1616. 21 390 reflections measured, of which 8302 were unique (R_{int} = 0.0362). Final R_1 = 0.0333 and wR_2 = 0.0725 for 6396 observed reflections with $I > 2\sigma(I)$.

OLED fabrication and measurements

The pre-cleaned ITO glass substrates were treated with ozone for 10 min. For the monochromatic OLEDs made by vacuum deposition, the NPB layer was deposited on a 10 nm MoO_3 thin layer pre-coated on the ITO glass. The Ir phosphor and CBP host were co-evaporated to form the emitting layer. Successively, BCP and Alq_3 or TPBi, LiF and Al were evaporated at a base pressure of less than 10^{-6} Torr. For the monochromatic and white-emitting PLEDs prepared by the solution process, the PEDOT:PSS and the emissive layers (chlorobenzene was used as the solvent) were spin-coated in sequence onto the pre-cleaned and UV–ozone treated ITO substrate and then annealed at 120 °C (45 min in air) and 100 °C (30 min in the glove box), respectively, followed by thermal evaporation of Ba (4 nm) and Al (100 nm) in a vacuum chamber at a base pressure of less than 3×10^{-6} Torr. All of the functional layers for the three-element WOLEDs made by vacuum deposition were stacked together by evaporation of the corresponding materials under a base pressure of less than 10^{-6} Torr. The EL spectra and CIE coordinates were measured with a PR650 Spectra colorimeter. The *L–V–J* curves of the devices were recorded using a Keithley 2400/2000 source meter and the luminance was measured using a PR650 SpectraScan spectrometer. All of the experiments and measurements were carried out at room temperature under ambient conditions.

Computational details

Calculations were performed with Gaussian 09 (ref. 36) at the Université de Sherbrooke and Mammouth super computer. The structures were optimized by DFT[37] (ground state) and TD-DFT[38] (excited state) B3LYP[39]/genecp calculations. For genecp basis sets, 6-31g*[40] were placed on C, H, N, O, F atoms, VDZ (valence double zeta) with SBKJC[41] effective core potentials were placed on Ir. The calculated absorption spectra and related MO contributions were obtained from gaussSum 2.1.[42]

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References

1 (*a*) M. A. Baldo, S. Lamansky, P. E. Burrows, M. E. Thompson and S. R. Forrest, *Appl. Phys. Lett.*, 1999, **75**, 4; (*b*) M. A. Baldo, M. E. Thompson and S. R. Forrest, *Nature*, 2000, **403**, 750.

2 (*a*) W.-Y. Wong, C.-L. Ho, Z. Q. Gao, B. X. Mi, C. H. Chen, K.-W. Cheah and Z.-Y. Lin, *Angew. Chem., Int. Ed.*, 2006, **45**, 7800; (*b*) Y. M. You and S. Y. Park, *Dalton Trans.*, 2009, 1267; (*c*) E. Baranoff, J. H. Yum, M. Graetzel and M. K. Nazeeruddin, *J. Org. Chem.*, 2009, **694**, 2661; (*d*) Y. Chi and P. T. Chou, *Chem. Soc. Rev.*, 2010, **39**, 638; (*e*) P.-T. Chou and Y. Chi, *Chem.–Eur. J.*, 2007, **13**, 380; (*f*) R. C. Evans, P. Douglas and C. J. Winscom, *Coord. Chem. Rev.*, 2006, **250**, 2093.

3 (*a*) C. Ulbricht, B. Beyer, C. Friebe, A. Winter and U. S. Schubert, *Adv. Mater.*, 2009, **21**, 4418; (*b*) W.-Y. Wong and C.-L. Ho, *Coord. Chem. Rev.*, 2009, **253**, 1709; (*c*) W.-Y. Wong and C.-L. Ho, *J. Mater. Chem.*, 2009, **19**, 4457; (*d*) G. Zhou, W.-Y. Wong and X. Yang, *Chem.–Asian J.*, 2011, **6**, 1706.

4 G. J. Zhou, C.-L. Ho, W.-Y. Wong, Q. Wang, D. G. Ma, L. X. Wang, Z. Y. Lin, T. B. Marder and A. Beeby, *Adv. Funct. Mater.*, 2008, **18**, 499.

5 (*a*) G. J. Zhou, W.-Y. Wong, B. Yao, Z. Y. Xie and L. X. Wang, *Angew. Chem., Int. Ed.*, 2007, **46**, 1149; (*b*) W. C. Wu, H. C. Yeh, L. H. Chan and C. T. Chen, *Adv. Mater.*, 2002, **14**, 1072; (*c*) X. M. Yu, H. S. Kwok, W.-Y. Wong and G. J. Zhou, *Chem. Mater.*, 2006, **18**, 5097.

6 (*a*) M. Nomuraa, Y. Shibasaki, M. Ueda, K. Tugita, M. Ichikawa and Y. Taniguchi, *Synth. Met.*, 2005, **151**, 261; (*b*) J. N. Wilking, B. Hsieh and G. A. Arbuckle-Keil, *Synth. Met.*, 2005, **149**, 63; (*c*) W.-Y. Wong, G.-J. Zhou, X. M. Yu, H.-S. Kwok and B. Z. Tang, *Adv. Funct. Mater.*, 2006, **16**, 838.

7 (*a*) S.-G. Jung, Y.-J. Kang, H.-S. Kim, Y.-H. Kim, C.-L. Lee, J.-J. Kim, S.-K. Lee and S.-K. Kwon, *Eur. J. Inorg. Chem.*, 2004, 3415; (*b*) M. S. Lowry, J. I. Goldsmith, J. D. Slinker, R. Rohl, R. A. Pascal, Jr, G. G. Malliaras and S. Bernhard, *Chem. Mater.*, 2005, **17**, 5712; (*c*) M. L. Xu, T. L. Li, W. L. Li, Z. R. Hong, Z. W. An and Q. Zhou, *Thin Solid Films*, 2006, **497**, 239.

8 A. Tsuboyama, H. Iwawaki, M. Furugori, T. Mukaide, J. Kamatani, S. Igawa, T. Moriyama, S. Miura, T. Takiguchi, S. Okada, M. Hoshino and K. Ueno, *J. Am. Chem. Soc.*, 2003, **125**, 12971.

9 S. Lamansky, P. Djurovich, D. Murphy, F. Abdel-Razzaq, H. E. Lee, C. Adachi, P. E. Burrows, S. R. Forrest and M. E. Thompson, *J. Am. Chem. Soc.*, 2001, **123**, 4304.

10 (*a*) A. P. Wilde, K. A. King and R. J. Watts, *J. Phys. Chem.*, 1991, **95**, 629; (*b*) S. Sprouse, K. A. King, P. J. Spellane and R. J. Watts, *J. Am. Chem. Soc.*, 1984, **106**, 6647.

11 X.-M. Liu, J. Xu, X. Lu and C. He, *Org. Lett.*, 2005, **7**, 2829.

12 C.-L. Ho, W.-Y. Wong, Q. Wang, D. G. Ma, L. X. Wang and Z. Y. Lin, *Adv. Funct. Mater.*, 2008, **18**, 928.

13 X. Yang, Y. Zhao, X. Zhang, R. Li, J. Dang, Y. Li, G. Zhou, Z. Wu, D. Ma, W.-Y. Wong, X. Zhao, A. Ren, L. Wang and X. Hou, *J. Mater. Chem.*, 2012, **22**, 7136.

14 C.-L. Ho, W.-Y. Wong, G. J. Zhou, B. Yao, Z. Y. Xie and L. X. Wang, *Adv. Funct. Mater.*, 2007, **17**, 2925.

15 R. C. Evans, P. Douglas and C. J. Winscom, *Coord. Chem. Rev.*, 2006, **250**, 2093.

16 M. A. Baldo, C. Adachi and S. R. Forrest, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2000, **62**, 10967.

17 (*a*) X.-N. Li, Z.-J. Wu, Z.-J. Si, H.-J. Zhang, L. Zhou and X.-J. Liu, *Inorg. Chem.*, 2009, **48**, 7740; (*b*) K. Chen, C.-H. Yang, Y. Chi, C.-S. Liu, C.-H. Chang, C.-C. Chen, C.-C. Wu, M.-W. Chung, Y.-M. Cheng, G.-H. Lee and P.-T. Chou, *Chem.–Eur. J.*, 2010, **16**, 4315; (*c*) Y. Wu, S.-X. Wu, H.-B. Li, Y. Geng and Z.-M. Su, *Dalton Trans.*, 2011, **40**, 4480; (*d*) S.-J. Liu, N.-N. Song, J.-X. Wang, Y.-Q. Huang, Q. Zhao, X.-M. Liu, S. Suna and W. Huang, *Phys. Chem. Chem. Phys.*, 2011, **13**, 18497.

18 (*a*) M. K. Nazeeruddin, R. T. Wegh, Z. Zhou, C. Klein, Q. Wang, F. De Angelis, S. Fantacci and M. Gratzel, *Inorg. Chem.*, 2006, **45**, 9245; (*b*) S. Bettington, M. Tavasli, M. R. Bryce, A. Beeby, H. Al-Attar and A. P. Monkman, *Chem.–Eur. J.*, 2007, **13**, 1423.

19 V. V. Grushin, N. Herron, D. D. LeCloux, W. J. Marshall, V. A. Petrov and Y. Wang, *Chem. Commun.*, 2001, 1494.

20 S. O. Jung, Y. Kang, H.-S. Kim, Y.-H. Kim, K. Yang and S.-K. Kwon, *Bull.Korean Chem. Soc.*, 2003, **24**, 1521.

21 D. Wang, J. Wang, H.-L. Fan, H.-F. Huang, Z.-Z. Chu, X.-C. Gao and D.-C. Zou, *Inorg. Chim. Acta*, 2011, **370**, 340.

22 T. Liu, B.-H. Xia, X. Zhou, Q.-C. Zheng, Q.-J. Pan and H.-X. Zhang, *Theor. Chem. Acc.*, 2008, **121**, 155.

23 S. Seki, Y. Sawada and T. Nishide, *Thin Solid Films*, 2001, **388**, 22.

24 S. C. Lo, N. A. H. Male, J. P. J. Markham, S. W. Magennis, P. L. Burn, O. V. Salata and I. D. W. Samuel, *Adv. Mater.*, 2002, **14**, 975.

25 T.-Y. Chu and O.-K. Song, *Appl. Phys. Lett.*, 2007, **90**, 203512.

26 M. L. Xu, T. L. Li, W. L. Li, Z. R. Hong, Z. W. An and Q. Zhou, *Thin Solid Films*, 2006, **497**, 239.

27 E. A. Plummer, A. van Dijken, H. W. Hofstraat, L. De. Cola and K. Brundwardner, *Adv. Funct. Mater.*, 2005, **15**, 281.

28 G. Zhou, W.-Y. Wong and S. Suo, *J. Photochem. Photobiol., C*, 2010, **11**, 133.

29 S. Chen, G. Tan, W.-Y. Wong and H.-S. Kwok, *Adv. Funct. Mater.*, 2011, **21**, 3785.

30 (*a*) Y. Sun, N. C. Giebink, H. Kanno, B. Ma, M. E. Thompson and S. R. Forrest, *Nature*, 2006, **440**, 908; (*b*) G. Schwartz, S. Reineke, T. C. Rosenow, K. Walzer and K. Leo, *Adv. Funct. Mater.*, 2009, **19**, 1319.

31 K. A. King, P. J. Spellane and R.-J. Watts, *J. Am. Chem. Soc.*, 1985, **107**, 1431.

32 (*a*) M. Thelakkat and H.-W. Schmidt, *Adv. Mater.*, 1998, **10**, 219; (*b*) R. S. Ashraf, M. Shahid, E. Klemm, M. Al-Ibrahim and S. Sensfuss, *Macromol. Rapid Commun.*, 2006, **27**, 1454.

33 SAINT, *Reference Manual*, Siemens Energy and Automation, Madison, WI, 1994–1996.

34 G. M. Sheldrick, *SADABS, Empirical Absorption Correction Program*, University of Göttingen, 1997.

35 G. M. Sheldrick, SHELXTL™, Reference Manual, version 5.1, Siemens, Madison, WI, 1997.

36 M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, V. G. Zakrzewski, J. A. Montgomery Jr, R. E. Stratmann, J. C. Burant, S. Dapprich, J. M. Millam, A. D. Daniels, K. N. Kudin, M. C. Strain, O. Farkas, J. Tomasi, V. Barone, M. Cossi, R. Cammi, B. Mennucci, C. Pomelli, C. Adamo, S. Clifford, J. Ochterski, G. A. Petersson, P. Y. Ayala, Q. Cui, K. Morokuma, D. K. Malick, A. D. Rabuck, K. Raghavachari, J. B. Foresman, J. Cioslowski, J. V. Ortiz, A. G. Baboul, B. B. Stefanov, G. Liu, A. Liashenko, P. Piskorz, I. Komaromi, R. Gomperts, R. L. Martin, D. J. Fox, T. Keith, M. A. Al-Laham, C. Y. Peng, A. Nanayakkara, C. Gonzalez, M. Challacombe, P. M. W. Gill, B. G. Johnson, W. Chen, M. W. Wong, J. L. Andres, M. Head-Gordon, E. S. Replogle and J. A. Pople, *Gaussian 98 (Revision A.6)*, Gaussian, Inc., Pittsburgh PA, 1998.

37 (*a*) P. Hohenberg and W. Kohn, *Phys. Rev.*, 1964, **136**, B864; (*b*) W. Kohn and L. J. Sham, *Phys. Rev.*, 1965, **140**, A1133; (*c*) *The Challenge of d and f Electrons*, ed. D. R. Salahub and M. C. Zerner, ACS, Washington, D.C., 1989; (*d*) R. G. Parr and W. Yang, *Density-functional Theory of Atoms and Molecules*, Oxford University Press, Oxford, 1989.

38 (*a*) R. E. Stratmann, G. E. Scuseria and M. J. Frisch, *J. Chem. Phys.*, 1998, **109**, 8218; (*b*) R. Bauernschmitt and R. Ahlrichs, *Chem. Phys. Lett.*, 1996, **256**, 454; (*c*) M. E. Casida, C. Jamorski, K. C. Casida and D. R. Salahub, *J. Chem. Phys.*, 1998, **108**, 4439.

39 (*a*) A. D. Becke, *J. Chem. Phys.*, 1993, **98**, 5648; (*b*) W. Y. Lee and R. G. Parr, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1988, 785; (*c*) B. Miehlich, A. Savin, H. Stoll and H. Preuss, *Chem. Phys. Lett.*, 1989, **157**, 200.

40 (*a*) R. Ditchfield, W. J. Hehre and J. A. Pople, *J. Chem. Phys.*, 1971, **54**, 724; (*b*) W. J. Hehre, R. Ditchfield and J. A. Pople, *J. Chem. Phys.*, 1972, **56**, 2257; (*c*) P. C. Hariharan and J. A. Pople, *Mol. Phys.*, 1974, **27**, 209; (*d*) M. S. Gordon, *Chem. Phys. Lett.*, 1980, **76**, 163; (*e*) P. C. Hariharan and J. A. Pople, *Theor. Chim. Acta*, 1973, **28**, 213; (*f*) J.-P. Blaudeau, M. P. McGrath, L. A. Curtiss and L. Radom, *J. Chem. Phys.*, 1997, **107**, 5016; (*g*) M. M. Francl, W. J. Pietro, W. J. Hehre, J. S. Binkley, D. J. DeFrees, J. A. Pople and M. S. Gordon, *J. Chem. Phys.*, 1982, **77**, 3654; (*h*) R. C. Binning Jr and L. A. Curtiss, *J. Comp. Chem.*, 1990, **11**, 1206; (*i*) V. A. Rassolov, J. A. Pople, M. A. Ratner and T. L. Windus, *J. Chem. Phys.*, 1998, **109**, 1223; (*j*) V. A. Rassolov, M. A. Ratner, J. A. Pople, P. C. Redfern and L. A. Curtiss, *J. Comp. Chem.*, 2001, **22**, 976.

41 (*a*) H. He, J. S. Binkley, J. A. Pople and W. J. Hehre, *J. Am. Chem. Soc.*, 1980, **102**, 939; (*b*) W. J. Stevens, H. Basch and M. Krauss, *J. Chem. Phys.*, 1984, **81**, 6026; (*c*) W. J. Stevens, M. Krauss, H. Basch and P. G. Jasien, *Can. J. Chem.*, 1992, **70**, 612; (*d*) T. R. Cundari and W. J. Stevens, *J. Chem. Phys.*, 1993, **98**, 5555.

42 N. M. O'Boyle, A. L. Tenderholt and K. M. Langner, *J. Comp. Chem.*, 2008, **29**, 839.